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An Idle Speed Controller using Analytically Developed Fuzzy Logic Control Law

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ABSTRACT

Fuzzy control is based on either expert knowledge or experimental data and, therefore, it possesses intrinsic qualities like robustness and ease of implementation. The mathematical modeling for fuzzy control systems has been attempted, but until now many models that have been developed do not extend beyond the application for which they were developed. A general class of fuzzy linguistic control algorithms that can be formulated analytically and can capture the nonlinear aspect of a given fuzzy control scheme has been formulated using interpolating functions. The interpolating functions map the process error and its rate or it's a cumulative sum into control action. The systematic approach makes it more desirable to be used in the control of nonlinear dynamics systems. The analytical method developed in previous work conducted by Langari (1992) and modified by the authors is employed in this practical example to design a controller for an idle speed control system. This study aims to show that the analytically developed fuzzy control law can be used in practical application such as the idle speed control problem provided that certain modifications and assumptions are taken into consideration in the derivation and the application of the control law.

The following sections present a discussion of the control law, in addition to the design process and the simulation results of the idle speed control system.

Analytically Developed Fuzzy Control Law For An Idle Speed Control System

Introduction

Fuzzy logic controllers (FLC) have been used extensively as an alternative to other control methods, such as PID control, adaptive control, observer-based control, and supervisory control. FLC is best suited for the control of nonlinear dynamic systems. Three methods of designing a fuzzy logic control are often used in one form or another. The three model or methods are the Mamdani Fuzzy Logic Controller (MFLC), the Takagi-Sugeno-Kang (TSK) Model, and the Simple Additive Model

(SAM). For in depth discussion of the three fuzzy models the reader is referred to (Ross 1996; Yen and Langari 1999; Passino and Yurkovich 1998).

The design process usually involves formulating the rules, which is a part of the knowledge base. In order to provide a smooth interface to the numerical process variable and set points, membership functions defining the linguistic terms are required. A fuzzifier determines the membership degrees of the controller input values in the If part of the rules. The inference mechanism combines the information with the knowledge stored in the rules and determines the outputs of the rule-based system. The output usually is a fuzzy set, although it can be a crisp set or singleton. Defuzzification is needed to calculate the overall value and produce a crisp controller output.

The reasoning mechanisms of a fuzzy controller with all of its reasoning actions are usually applied in the design process. The choices of universe of discourse and its division into fuzzy sets are dictated by the system. The membership function shape and the defuzzification methods can be chosen from many methods that are available. However, there is no systematic way of designing a fuzzy controller. Previous works used different methods for different systems. Shigeyasu et al. (1991) used knowledge acquired from PD control to design the fuzzy logic controller. A PID type fuzzy controller using parameter adaptation method to produce a parameter adaptive fuzzy controller was designed by Wu and Mizumoto, (1996).

Ostergaard (1977) expressed the fuzzy sets in a functional form, with each form is associated with a linguistic label. In order to systematically set a guideline for the design process, certain defuzzification, aggregation, and membership function shapes are assumed. For example, using the product instead of Max-Min in the aggregation steps and center of area method (COA) for defuzzification process, a functional fuzzy logic control algorithm can be developed. A general functional control algorithm has been attempted (Langari, 1992; Wang, 1997; Passino and Yurkovich, 1998). Some attempts have taken the form of empirically based comparative studies or analytical studies. Langari (1992) developed a nonlinear formulation of fuzzy linguistic control algorithms. The analytical model developed by Langari (1992) is applied to the idle speed control problem with suitable assumptions and modifications. Idle speed control problem has been an active research subject lately because of the desire to produce fuel-efficient vehicles and to satisfy the stringent environmental regulation of low pollutant emissions. The model is applied with some modification to fit the problem requirements. Comparing the controller performance with other control methods previously used assesses the validation of the model. A typical Proportional plus Derivative (PD) fuzzy based control system is shown in Figure 1.

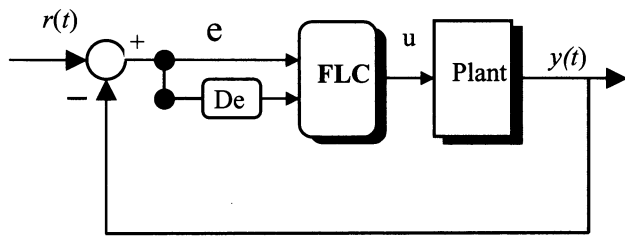


Figure 1. A PD-Type Fuzzy Based Control System.

The inputs to the fuzzy logic controller in Figure 1 are the error and the cumulative sum of the error or the integral of the error. The error is defined as the difference between a set reference value and the measured value, i.e. $(r(t) - y(t))$. The output of the fuzzy controller $u(t)$ is the input to the plant or the process. The fuzzy logic controller control rules can be stated in general form as:

$R_{j,l}$: If $e(t)$ is A_j and $De(t)$ is B_l Then $u(t)$ is $C_{j,l}$

where $De(t)$ stand for $\dot{e}$ in the case of PD-type fuzzy logic controller or $\int e \, d\tau$ for the PI-type fuzzy logic controller. The

symbols A_j, B_l are the linguistic terms such as *small, high, short, medium*, etc. The linguistic variables are subsets of the universe of discourse of the input variable $e(t)$ and $De(t)$,

respectively. Figure 2 shows the universe of discourse of the error (e), its rate, and output (u). The symbols $NB, NM, NS, Z, PS, PM, PB$ in Figure 2 are the linguistic variables that are given by the subsets A_j , and they stand for *Negative Big, Medium, and Small; Positive Big, Medium, and Small*, and Z stands for *Zero*. The fuzzy set constructed using the left and right parameterization (LR) is given by Equation (1).

$$\mu_A(u) = \begin{cases} L(\frac{u_0 - u}{\alpha}) \dots \dots \dots \text{if } u \leq u_0 \\ R(\frac{u - u_0}{\beta}) \dots \dots \dots \text{if } u > u_0 \end{cases} \quad (1)$$

In most practical applications in which fuzzy logic control is used the fuzzy sets employed are symmetrical fuzzy sets. The left and right parameterization functions for symmetrical fuzzy sets are equal and $\alpha = \beta$. Also, for symmetrical fuzzy set the left or right parametric function takes any of the forms in Equation (2).

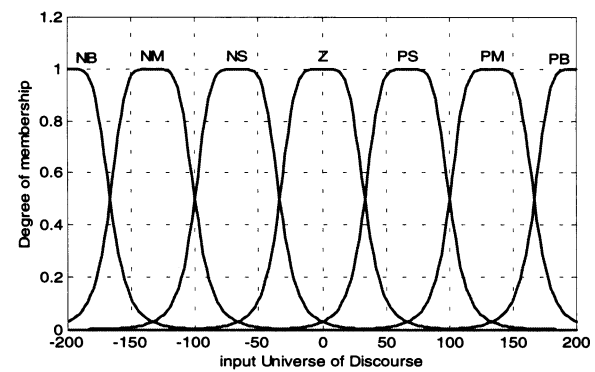


Figure 2. Linguistic Values Represented by Membership Functions.

$$L(x) = \begin{cases} \max(0, 1 - |x|^p) \\ e^{(-|x|^p)} \\ \frac{1}{1 + |x|^p} \end{cases} \quad (2)$$

In the case where $L(x) = \max(0, 1 - |x|^p)$ is used, where the Max operator takes on the maximum value in the brackets the resulting membership function is the well-known triangular membership function. The argument x in this case can take on the values bracketed for the R or L function, and α and β are defined as the inverse of the slopes m_L and m_R , respectively. A detailed description of (LR) parameterization can be seen in Langari (1992). The control law obtained using (LR)

parameterization is derived and given in Equation (3). A detailed control law derivation is omitted for brevity purposes.

$$u(t) = [(1 + m_j E_j) \Delta K_j (e(t) - E_j) + K_j e(t)] + [(1 + m_l DE_l) \Delta K_l (de(t) - DE_l) + K_l de(t)] \quad (3)$$

where

$$m_j = \frac{1}{\alpha_{j+1}} = \frac{1}{\beta_j} \quad m_l = \frac{1}{\alpha_{l+1}} = \frac{1}{\beta_l}$$

$$K_{j+1} = K_j + \Delta K_j \quad K_{l+1} = K_l + \Delta K_l$$

$$E_{j+1} = E_j + \frac{1}{m_j} \quad DE_{l+1} = DE_l + \frac{1}{m_l}$$

$$U_j = K_j E_j + K_l DE_l \quad (4)$$

The derived control law is then used in idle speed control system. In the following section, a brief introduction to the idle speed control problem is discussed and the problem statement is presented.

Idle Speed Control

The evolution of engine control from mechanical to electronic and from open loop to closed loop is seen in the improvement in both hardware and software. The use of electronics improves sensing accuracy, actuation capability, and flexibility in control law design. Applying the modern control and estimation theory requires a dynamic engine model that describes the effects of control variables on the engine outputs. Based on the model, a microprocessor can be developed and real time implementation can be achieved. The objective for idle speed control is to maintain the engine idle speed as close to the desired set idle speed as possible. Load torque disturbances such as turning on and air conditioning, power steering, and transmission shift can cause engine speed to fluctuate widely around the idle speed. Aging of automotive and variation in fuel efficiency, emissions and automotive vibrations can also effects the idle speed set point. In this work the control law developed in Equation (3) is applied to control the idle speed of a spark ignition engine. An analytically develop fuzzy control law makes the design of a fuzzy logic controller for this purpose much easier and more attractive than traditional approaches. The experts' knowledge or the data collected from on-line experimentation can be used to derive the rule-base fuzzy logic control law. The advantage of using the analytical model is the simplification of the overall design process mentioned earlier. The biggest drawback of using fuzzy logic controllers is that there is no

systematic way of going about designing such a control law. However, using the developed model the design process can be clearly stated and outlined and the controller can be designed.

Previous work related to the present work could be seen in Boverie et al. (1997), where experimental engine data were used to develop the rule-based control system. The obtained results were very comparable to those obtained using PID and LQ optimal control. A neural network based procedure developed by (Feldkamp and Puskorius, 1993) was used to train fuzzy system to control vehicle idle speed; the results were comparable to linear controller schemes and in some cases performed better than a linear controller.

In the following section the problem statement for the idle speed control is presented and the analytical model is used to develop the control law.

Idle Speed Fuzzy Controller Using the Developed Model

To achieve a good idle speed control, several requirements must be considered. Optimization of the drivability and comfort, as well as the minimization of fuel consumption and pollutant emissions are factors that can fall into the requirements category. Stiffening of the rules against pollutant emissions has led to a reduction of the idle speed set point. This reduction in turn has great effects on the stability of the engine at that lowered set speed. Engine process at idle speed can be analyzed in terms of inputs and outputs. The inputs control variables are the fuel injection pulse width, spark advance, and throttle angle, which determine the air mass flow entering the cylinders. The measured outputs are the manifold pressure and engine speed. The idle speed engine controller is usually designed to control simultaneously the amount of air at the inlet of the engine by controlling the opening of the throttle valve and the spark advance which provide the ignitions to power stroke. In this work a conventional PID controller controls the spark advance part of the control. The throttle angle opening and closing can be used to control the amount of air mass flow into the cylinders. The fuzzy controller designed here is then the one that controls the air mass flow. The main inputs to the fuzzy controller are the engine speed error (e) and the rate of the error (De).

Figure 3 shows the general structure for the proposed fuzzy controller. The idle speed controller is switched on provided that any one of the following conditions occur:

1. The engine speed falls below threshold value, which is preselected fixed value, i.e., $N \leq N_0$.
2. The rate of the measured engine speed or the rate of the engine speed error is positive, i.e., $N(k) - N(k-1) > 0$.
3. The accelerator pedal is released, i.e., $N(k) \ll N(k-1)$.

The idle speed controller must be applied successfully when the vehicle is either in motion or stationary. If the vehicle is in motion, then the dynamics of the engine and the vehicle should be considered, while in the stationary state only the engine

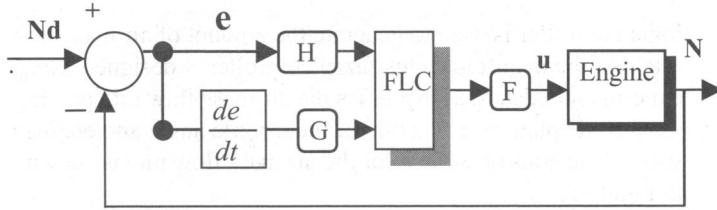


Figure 3. Idle Speed Control system Diagram. H and G and F are scaling factors for (e) , (De) , and u respectively.

dynamics need to be considered. In designing the proposed fuzzy controller, the inputs and outputs are identified and the range of operation for each is used to define the number of membership functions to be used. The universe of discourse for the error is divided into seven fuzzy sets ($A_j, j=1...7$).

Each fuzzy set is given linguistic value ranging from Negative Big (**NB**) to Positive Big (**PB**) with Zero (**Z**) representing the fuzzy set around or close to the desired set speed. The same is done for the rate of the error (De), only the fuzzy sets in this case are represented by ($B_l, l=1...7$). The output universe of discourse is divided into fuzzy sets of the form $C_{j,l}$. Figure 4 shows the partitions of the universes of discourse. In 8 the **NB** corresponds to A_{-3}, B_{-3}, C_{-3} , **NM** corresponds to A_{-2}, B_{-2}, C_{-2} , **NS** corresponds to A_{-1}, B_{-1}, C_{-1} , **Z** corresponds to A_0, B_0, C_0 , **PS** corresponds to A_1, B_1, C_1 , **PM** corresponds to A_2, B_2, C_2 , and **PB** corresponds to A_3, B_3, C_3 . The parameter β is the value at which the membership function or fuzzy set is 1 i.e., $E_j, DE_l, U_{j,l}, j = l = -3, \dots, 3$.

The control rules are expressed as:

If e is A_j and De is B_l Then u is C_{jl}

The same rule can be expressed in terms of the linguistic values as:

If e is **PB** and De is **NM** Then u is **PS**

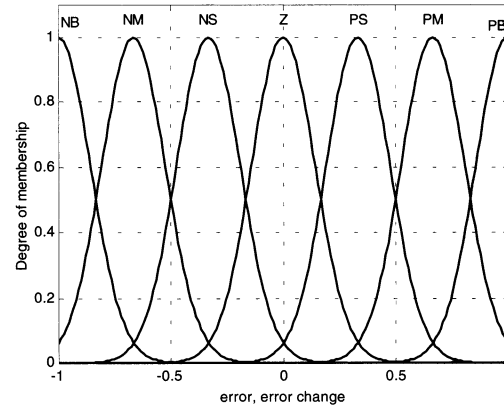


Figure 4. The Universe of Discourse for the Inputs Error (e), Rate of the Error (De), and Output (u).

At most four rules can be fired at anytime, i.e., given the error, and change of the error then at most four combination of the error and its change can be obtained to give the control action u . This is because of the overlapping of the membership functions. The rules fired at any given are:

If e is A_j and De is B_l Then u is C_{jl}

If e is A_j and De is B_{l+1} Then u is $C_{j,l+1}$

If e is A_{j+1} and De is B_l Then u is $C_{j+1,l}$

If e is A_{j+1} and De is B_{l+1} Then u is $C_{j+1,l+1}$

From experimentally obtained data, one can derive the fuzzy rules. Many methods than can be used for doing that such as fuzzy clustering. The fuzzy clustering using gradient method is employed in this study. In using fuzzy clustering it is always easier and more practical to do it using TSK fuzzy model rather than Mamdani type. The mathematical formulation of TSK makes it more desirable for such application. If Mamdani type fuzzy logic controller is desired then clustering can be accomplished using TSK and the resulted TSK fuzzy model can be converted into Mamdani type fuzzy system. This process to derive the fuzzy rules for the analytical model was employed in this study.

A fuzzy associate memory (FAM) table is then created for all of the rules. Table 1 shows the rules structures for the proposed fuzzy controller. The set speed is chosen to be 750 rpm.

The scaling factors for the error, its change, and the output are determined from the normalization of their respective universe of discourse. The formulas in Equation (5) are used in normalizing the error and the error change.

				E					
		u	PB	PM	PS	Z	NS	NM	NB

Table 2. FAM Table. Entries are the control action u .

$$\begin{aligned}
 E_N &= K_1 + K_2 E \\
 K_1 &= \frac{E_L + E_R}{E_R - E_L} E_{CL} \\
 K_2 &= \frac{2}{E_R - E_L} E_{CL}
 \end{aligned} \quad (5)$$

This transforms the universe of discourse of the error from the range defined by $[E_L, E_R]$ to the normalized range defined by $[-E_{CL}, E_{CL}]$. The normalized universe of discourse E_N resulted from this transformation is then defined on the interval $[-E_{CL}, E_{CL}]$. Applying K_N to the universe of discourse transforms it into $[-1, 1]$. The fuzzy control action based on the developed control law along with the design procedure outlined can be found by different methods. Depending on the method chosen and the membership functions being either symmetric or nonsymmetric. In this study where the error and the error change are used as the inputs to the controller with symmetric membership functions and product-max implication the control action is given in equation (6)

$$\begin{aligned}
 u &= \frac{(DE_{l+1} - e) \cdot (E_{j+1} - \dot{e}) \cdot C_{jl} + (e - DE_l) \cdot (E_{j+1} - \dot{e}) \cdot C_{(j+1)l}}{D} \\
 &+ \frac{(DE_{l+1} - e) \cdot (\dot{e} - E_j) \cdot C_{j(l+1)} + (e - DE_l) \cdot (\dot{e} - E_j) \cdot C_{jl}}{D}
 \end{aligned} \quad (6)$$

where $D = (DE_{l+1} - DE_l) \cdot (E_{j+1} - E_j)$.

In terms of the degree of membership functions the same control action in Equation (6) can be expressed as:

$$u = (\alpha_j \cdot \beta_l + \alpha_{j+1} \cdot \beta_{l+1}) \cdot C_{jl} + \alpha_{j+1} \cdot \beta_j \cdot C_{(j+1)l} + \alpha_j \cdot \beta_{l+1} \cdot C_{j(l+1)}$$

Simulation Results and Discussion

In order to test the fuzzy controller performance one need to either conduct an experiment or simulate the controller algorithm by writing a computer program or using commercially available control system software. For

simulation purpose a mathematical model for the different subsystems of the controlled process is needed. In the case of idle speed, throttle, intake manifold, air/fuel flow dynamics, compression stroke, and torque generation and acceleration subsystems need to be mathematically modeled. The mathematical model developed by Grossley and Cook, (1991) is used in this study. It is worth noting that an embedded fuzz logic controller is used to generate the amount of air mass flow rate into the manifold. This fuzzy controller is designed using experimental data that expresses the air mass flow rate passing the throttle plate as a function of the throttle angle and engine speed. The control surface for the air mass flow rate is shown in Figure 5.

The controller is designed using fuzzy c-mean clustering to derive the rules and the membership functions for the controller inputs. The output of the controller is expressed in the same format as TSK fuzzy model, i.e.,

$$\dot{m}_{ai} = k_o + k_1 \theta + k_2 N.$$

The simulation for the above controller is then conducted using SIMULINK with fuzzy logic toolbox and Matlab control toolbox. The SIMULINK model is shown in Figure 6.

The range of the engine speed is taken to be in the interval of $\{600, 3000\}$ rpm.

The air valve opening, which is expressed in percentages, is defined on the interval of $\{0, 100\}\%$, where the range 0 -15% throttle is what is known as closed throttle; 70-90% throttle indicates a wide open throttle; and any value in between is known as the part throttle situation.

The spark advance is assumed to be constant when the engine is idling, and it is chosen to be in the interval of $[0, 45]$ degree BTDC. A conventional proportional plus derivative plus integral (PID) controller controls the spark advance.

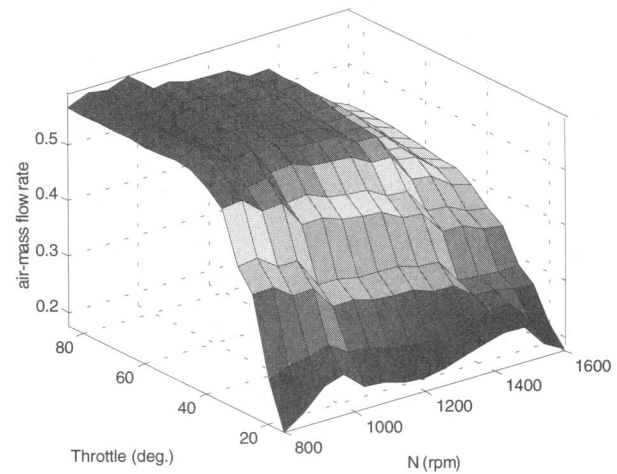


Figure 5. Air-Mass Flow Rate Control Surface Generated Using C-Mean Clustering.

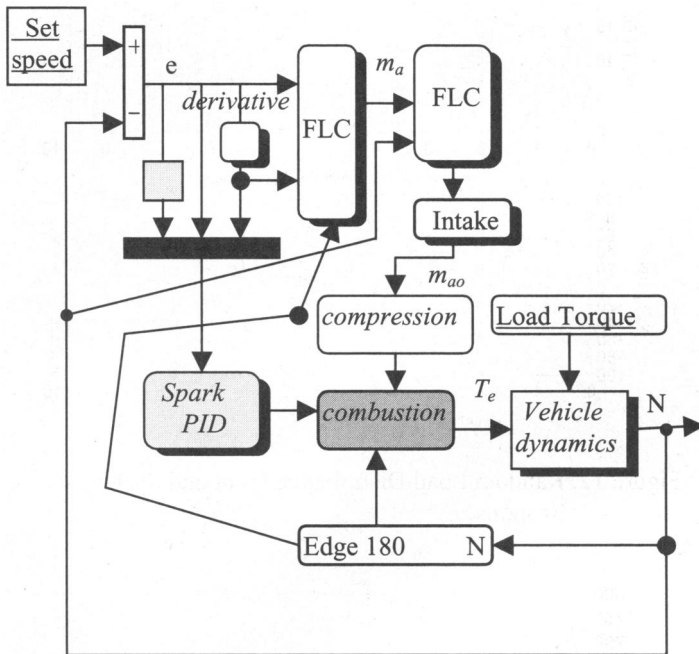


Figure 6. SIMULINK Simulation Model.

Figure 7 shows the engine response for positive load torque and throttle angle. The figure shows how the engine speed error goes to zero for different throttle load torque. The system responded well to perturbation and changing load torque.

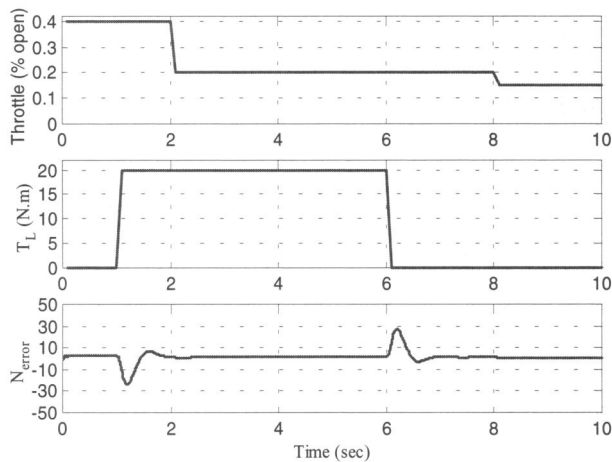


Figure 7. Engine Speed Error (N_{error}). Throttle Input (Top), Load Torque of 20 N.m. (Middle). Engine Speed Perturbation (Bottom).

Figure 8 is shown for acceleration, where again we see the controller was able to bring the transition speed after the acceleration pedal is off to a desired idle engine speed.

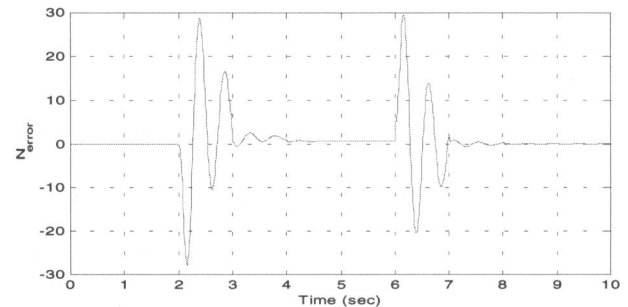


Figure 8. Acceleration and Deceleration and the Controller Action.

Figure 9 shows the load torque and the response for the speed throttle control. Again, the controller was able to bring the idle error to zero for changing load torque. The time it took the controller to do so is a little long and the overshoot is acceptable in this situation. The overall response is acceptable. Figure 10 shows the throttle input and the system response when a positive load constant torque of 20 rpm is applied. The controller was able to keep zero speed error and with very minimal initial overshoot.

Figure 11 shows the system response for applied load torque along with the desired idle speed. It can be seen that the steady state error is within ± 50 rpm. This result compares very well with other results obtained by other authors with a range of ± 150 rpm. One can see that the fuzzy controller was able to bring down the deviation to ± 10 rpm. The deviation between

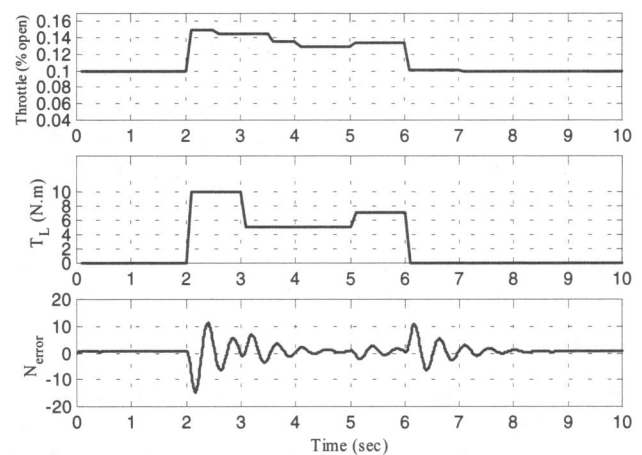


Figure 9. Idle Speed System Response. Throttle (top), Load Torque Input (Middle) and the System Response (Bottom).

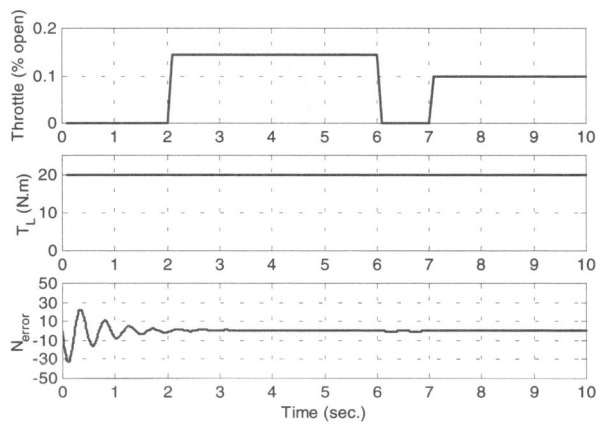


Figure 10. Idle Speed System Response for Throttle Input and Constant Load Torque.

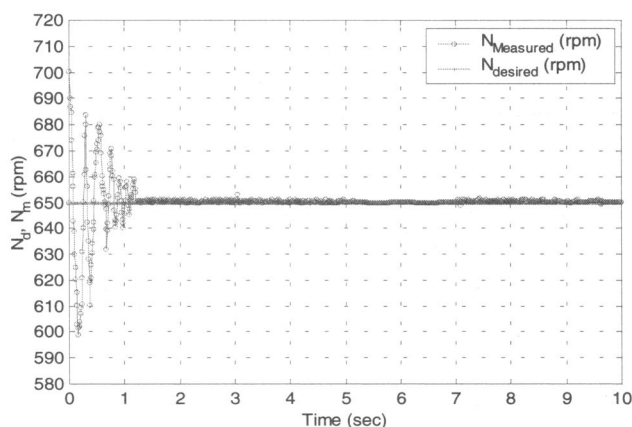


Figure 11. Desired and Measured Engine Speed when Load Is Applied.

the measured and set speed in Figure 11 is within small limit, which in the case of idle speed control is acceptable. Figure 12 shows the engine idle speed control for a random input. The engine speed perturbations are very small and are much better than the ones obtained with other control strategies.

Figure 13 shows the system step response. The figure shows that there is a minimal overshoot and short settling time with zero steady state error. The controller in this case over performed the presently used conventional PID controller.

Figure 14 is a comparison between the Linear Quadratic Gaussian (LQG) control scheme and the fuzzy logic control method used in this study. Figure 14 clearly shows that the performance of fuzzy logic controller is better than that of the LQG.

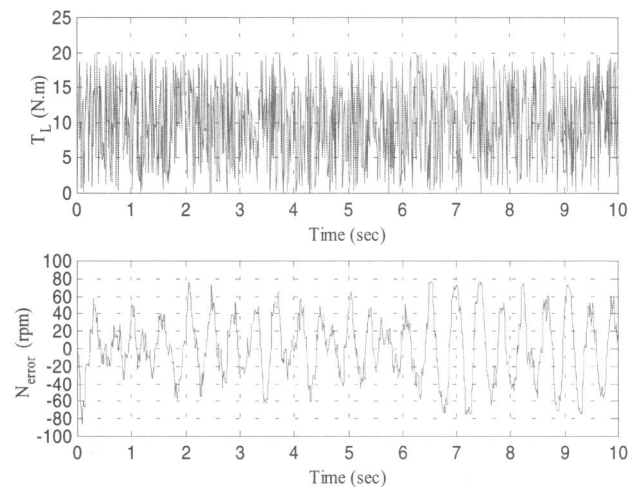


Figure 12. Random Load Disturbance Input and System Response.

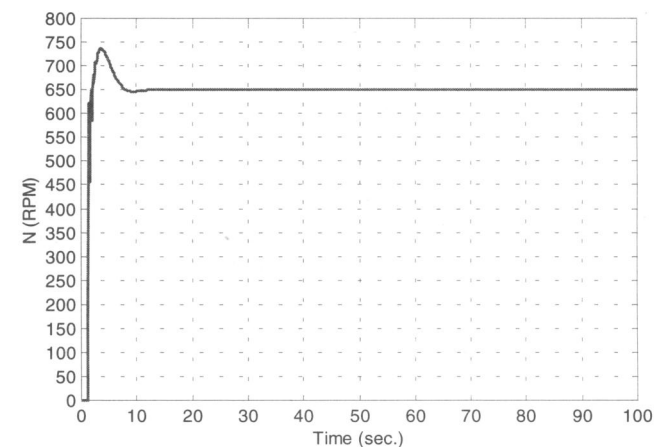


Figure 13. The System Step Response.

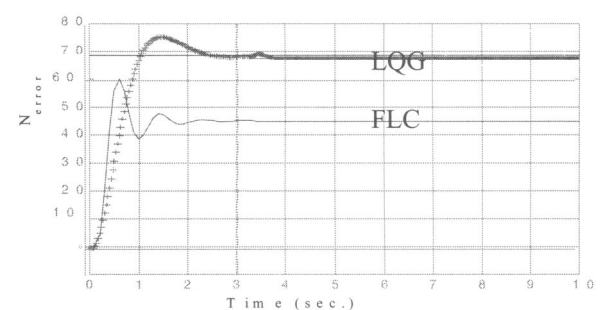


Figure 14. Comparison Between Linear Quadratic Gaussian(LQG)(+) and Fuzzy Logic Controller (Solid).

Concluding remarks

This study dealt with the design and implementation of the fuzzy logic controller using an analytical developed expression of the control law. The fuzzy logic controller was then used in the closed loop feedback control system to control the idle speed of spark ignition engine. The results obtained using the fuzzy logic designed in this study are shown to be comparable and, in some cases, better than conventional based control methods that are presently used or under investigation. The idle speed for the different operations of the vehicle was investigated and the effectiveness of the idle speed controller designed here is shown to be valid under those conditions. The idle speed error was reduced significantly. The steady state error was also reduced and in some cases it was eliminated. The design methodology used in this paper can be applied to other engine control systems or vehicle control systems, such as cruise control, air/fuel ratio control, and many other automotive-related applications. This study shows that one can design a fuzzy logic controller systematically if certain information about the plant and the inputs and output of such plant are known. The developed controller is economical in terms of cost and time spent in the design process and expedient in terms of prototyping and real time application.

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